DLD → Digital Logic Design 109.

⇒ Introductive Class

Syed Amasil (14 grade)

Ned university → Bachelor's → masters

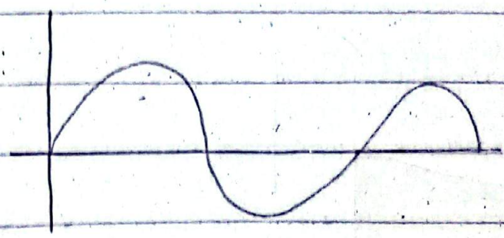
X ———— X ———— X

speed → fast
size → compact

analog devices → digital technologies

Analog Single → continuous
speed
کم زیادہ ہو سکتا ہے۔

Digital Single →



+5 volts represents logic high state
↳ logic 1

0 volts represents

25/02/2026

Number System

* Binary Number System Convertion:

$(\text{binary})_2$

2	86	
2	43	— 0
2	21	— 1
2	10	— 1
2	5	— 0
2	2	— 1
2	1	— 0

$(01101)_2$

$(1010110)_2$

Home Work: <u>Number System Convertion</u>

binary

2	56	
2	28	— 0
2	14	— 0
2	7	— 0
3	3	— 1
	1	— 0

(101000)

binary conversion.

JDK

NetBeans — IDE official website

Private → access → getter / setter

Java deep key point

W3School.

system.print.ln ("Hello World"):

next line

path set karna → JDK, installation.

OAK - java previous name

Java runtime environment → JRE

oktal to decimal

$(2374)_8 = (2\times8^3)+(3\times8^2)+($

Octal no. System:

A system composed of eight digits which are,
0 1 2 3 4 5 6 7

Octal to Decimal

$(2\,3\,7\,4)_8$ (place values 3 2 1 0; base eight)

$= (2\times 8^3) + (3\times 8^2) + (7\times 8^1) + (4\times 8^0)$

$= (2\times 512) + (3\times 64) + (7\times 8) + (4\times 1)$

$=$

$= (1276)_{10}$

Decimal to Octal

$\boxed{\begin{array}{l} 8^0=1 \\ 7^0=7 \\ a^0=1 \end{array}}$

$\frac{359}{8} \rightarrow 44.875$

$0.875 \times 8 = 7$

$\frac{44}{8} \rightarrow 5.5$

$0.5 \times 8 = 4$

$\frac{5}{8}$

$0.625 \rightarrow 0.625 \times 8 = 5$

$\boxed{547_8}$

اس وقت تک divide کرنا ہے جب تک 0
نہ آجائے۔

Octal to Binary:

represents. 3 bits

421	
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

000
010
100

Q. binary no. base 2

$(110011011)_2$

1.) 3 pairs bana lye

421	
110	6
011	3
011	3

Bina
$(110011011)_2 \Rightarrow (633)_8$

Hexa-decimal no. system:

Machine language uses hexa-decimal number system. (four bit group)

hexa-decimal uses 16 characters

8 4 2 1		decimal
0 0 0 0	0	0
0 0 0 1	1	1
0 0 1 0	2	2
0 0 1 1	3	3
0 1 0 0	4	4
0 1 0 1	5	5
0 1 1 0	6	6
0 1 1 1	7	7
1 0 0 0	8	8
1 0 0 1	9	9
1 0 1 0	A	10
1 0 1 1	B	11
1 1 0 0	C	12
1 1 0 1	D	13
1 1 1 0	E	14
1 1 1 1	F	15

Binary to Hexa-decimal

<u>0011111100010110 1001</u>

8421	
0011	3
1111	F
0001	1
0110	6
1001	9

$(0011111100010110 1)_2 \rightarrow (3F169)_{16}$

"This conversion is not possible"

Hexa-decimal to binary:

$(10A4)_{16} \rightarrow (\quad)_2$

8421	
0001	1
0000	0
1010	10(A)
0100	4

(~~000~~1 ~~0000~~ ~~0000~~ 0001)

$(0001000010100100)_2$

$(10A4)_{16} \rightarrow (0001000010100100)_2$

hexa-decimal to binary

Hexa-decimal to Decimal

hexa-deci → B
B → D

2 way process

a) $1C_{16} \longrightarrow 2^4 + 2^3 + 2^2$

$16 + 8 + 4$

28

0001 1100

$\cancel{0\times2^5} + \cdot\times2^4 + \times2^3 + c\times2^2 + 0\times2^1 + 0\times2^0$

$\Rightarrow \cancel{0\times2^6} + \cancel{0\times2^7} + \cancel{0\times2^8}$

$0\times2 +$

$\Rightarrow (1\times2^4) + (1\times2^3) + (1\times2^2)$

$= 16 + 8 + 4$

$= 28_{10}$

X ———— 8421 X ———— X

b.) $A85_{16}$

1010 1000 0101

$1\times2^7 + \cancel{0\times2^6} + \cancel{0\times2^5} + \cancel{0\times2^4} + \cancel{0\times2^3} + 1\times2^2 + \cancel{0\times2^1} + 1\times2^0$

$\cancel{0\times2^8} + 1\times2^9 + \cancel{0\times2^{10}} + 1\times2^{11}$

$= 2^7 + 2^9 + 2^2 + 2^0 + 2^{11}$

$= 2693_{10}$ Ans.

10/3/2026

Decimal to Hexa-decimal

650 → hexadecimal

$\frac{650}{16} \rightarrow 40.625 \rightarrow 0.625\times16 = 10$

$\frac{40}{16} \rightarrow 2.5 \rightarrow 0.5\times16 = 8$

$\frac{2}{16} \rightarrow 0.125 \rightarrow 0.125\times16 = 2$

$\cancel{\frac{0}{16}}$

2 8 10

Check answer is corect or not.

$(28A)_{16}$

16	650
16	40 — 10(A)
	2 — 8

$(28A)_{16}$

Octal to Binary

$(13)_8 \Rightarrow$ 3 bit

Binary

$$\begin{array}{cc} 1 & 3 \\ 421 & 421 \\ 001 & 011 \end{array} \Rightarrow (13)_8 \Rightarrow (001011)_2$$

001011

Decimal Faraction to Binary:

$0.3125 \times 2 = 0.625 \rightarrow 0$ 0101

$0.625 \times 2 = 1.25$ 1

$0.25 \times 2 = 0.5$ 0

$0.5 \times 2 = 1.00$ 1

0101

Binary Arithmatic

Addition:

$0 + 0 = 0$

$0 + 1 = 1$

$1 + 0 = 1$

$1 + 1 = 10$

exp:

$$\begin{array}{r} 011 \\ +\ 001 \\ \hline 100 \end{array} \qquad \begin{array}{r} 1101 \\ +\ 1010 \\ \hline 10111 \end{array}$$

$(100)_2$ $(10111)_2$

Subtraction:

$0 - 0 = 0$

$1 - 1 = 0$

$1 - 0 = 1$ 1101

$10 - 1 = 1$

exp:

$$\begin{array}{r} 0\cancel{1}01 \\ -\ 011 \\ \hline 010 \end{array} \qquad \begin{array}{r} 1\cancel{1}01 \\ -\ 0011 \\ \hline 1010 \end{array} \qquad \begin{array}{r} 0\cancel{1}0\cancel{1}01 \\ -\ 01010 \\ \hline 01011 \end{array}$$

$(010)_2$ $\cancel{(1010)_2}$ $(01011)_2$

The solution of this question is not possible.

Multiplication :

$0 \times 0 = 0.$
$0 \times 1 = 0$
$1 \times 0 = 0$
$1 \times 1 = 1$

multiplication exp home work hai.

decimal → binary

Divison

Complement of Binary Number

1's complement:

00000101   Find 2's complement.

11111010 → 1's complement

2's complement = 1's comple. + 1 (end)

$$\begin{array}{r} 11111010 \\ +\ 1 \\ \hline 11111011 \end{array}$$

Sign Bit:

00011001

Left-most bit

+ev ہو تو 0

-ev ہو تو 1

+25 , -25

Q-39 ~~ato~~ Express -39 as an 8-bit number

- 39

8421 8421
0011 1001

00111001

11000110   1s' comp

11000111   2s' comp

Floating Point Number

Flo point number → 32 bit sy represent kiye jaty

23 slide
tak study

Logical Gates

Not, AND, OR ⟶ Basic Gates

NAND, NOR ⟶ Universal Gates

XOR, XAND ⟶ Arithmatic Gates

X ——— X ——— X

// Not Gate

The input is low, the output is high.
The output is low, the input is high

A = input
X = output

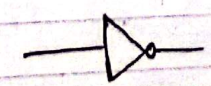
Input	Output
Low(0)	High(1)
High (1)	Low (0)

(NOT circuit)

$X = \bar{A}$

if x = 1 then A = 0
if x = 0 then A = 1

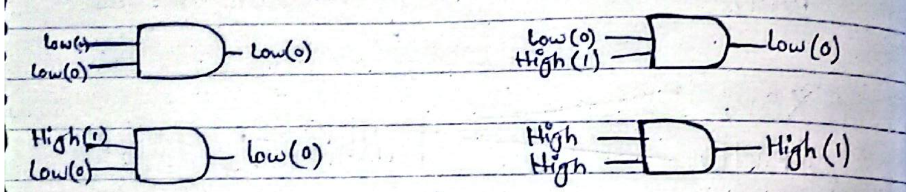
read as: "A bar"
: "not A"

A  $X = \bar{A}$

AND Gate : (required more than 2 inputs (A, B))

⇒ output X is High (1) only when A & B are high (1)

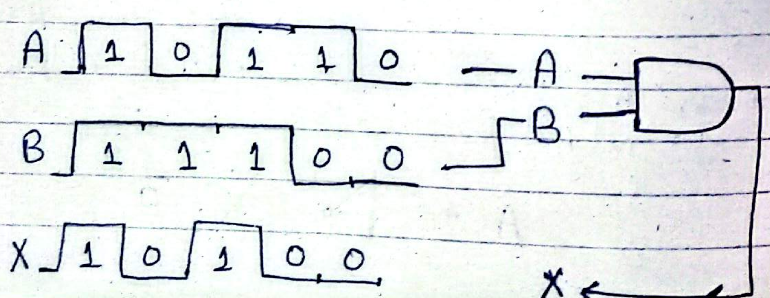
⇒ output X is low when either A or B, or when both A & B are low.



Truth table for 2 input AND Gate

Input		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

Inputs			Output
A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	1	1
1	1	0	0

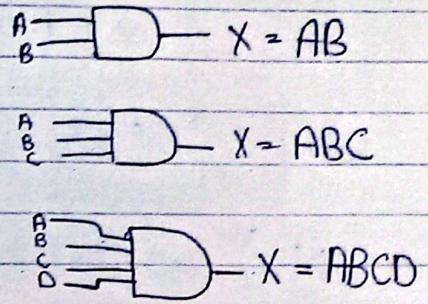


Bolean multiplication is the same as AND functi

X = AB

X = ABC

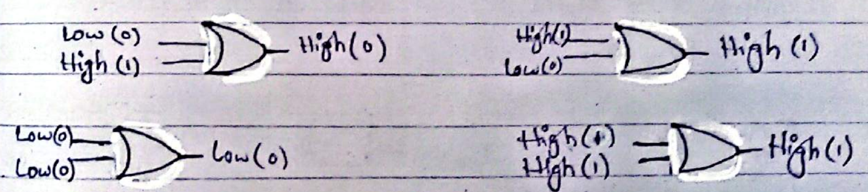
X = ABCD



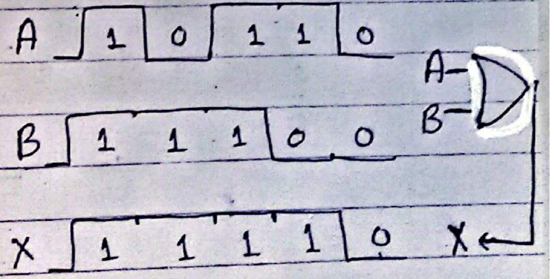
OR Gate (required more than 2 inputs)

⇒ output X is High when either A or B is Heigh or both A & B are High

⇒ output X is low when both A & B are low

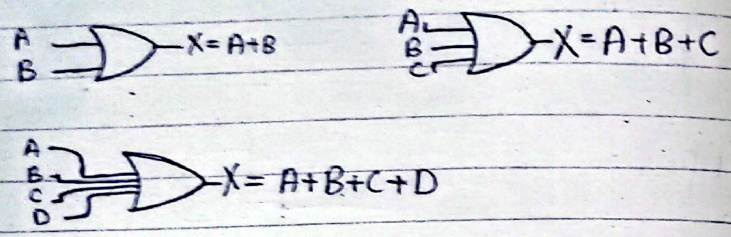


Input		Output
A	B	X
0	1	1
0	0	0
1	0	1
1	1	1



Boolean Addition is the same as OR functi.

$X = A + B$



Boolean expression for OR Gate with two three and four inputs.

X ——— X ——— X

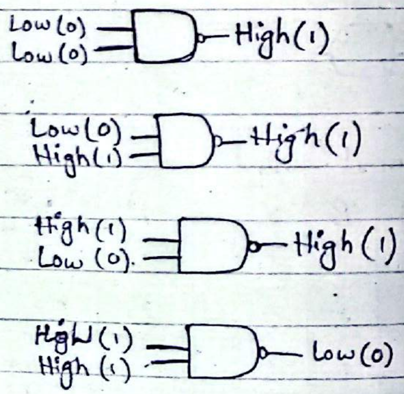
//NAND :
(opposite of OR)

جب input 0 ہوگا 0 ہوگا و output High (1) ہوگا

⇒ output X is low when input A & B are High.
⇒ output X is Hight when input A or B is low. or both are low.

truth table of NAND Gate

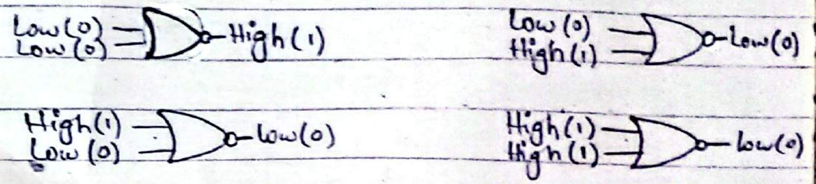
A	B	X
0	0	1
1	0	1
0	1	1
1	1	0



//NOR : (opposite of AND)

⇒ output X is low when either input A or B is High or both A & B are High.

⇒ X is only High when A & B are low



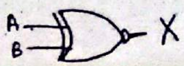
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

//XOR [Exclusive OR]

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

// XNOR

A	B	X
0	0	1
1	0	0
0	1	0
1	1	1



Boolean Algebra